

Changbo Wang

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EDUCATION

Northeastern University

Boston, MA

B.S. in Computer Science and Game Development | GPA: 3.67

Sep 2024 – May 2028

Relevant Coursework: Algorithms & Data; C++ Programming; Program Design and Implementation

TECHNICAL SKILLS

Languages & Core: Java, Python, C/C++, C#, JavaScript, TypeScript, SQL, HTML/CSS; OOP

Frameworks & Platforms: Spring Boot, Spring Data JPA, React, PostgreSQL, Unity, WordPress, Shopify

Tools & Testing: Git/GitHub, Docker Compose, JUnit, Flyway, Figma; debugging and unit testing

PROJECTS

CourseFlow — Course Planner (In Development)

Java, Spring Boot, React, PostgreSQL

- Developing a full-stack course planner with React, TypeScript, Spring Boot, and PostgreSQL for course search, section comparison, and schedule generation using clearly labeled demo seat data.
- Implemented backtracking with early conflict pruning and preference scoring; added JUnit tests for overlapping meetings, adjacent time boundaries, and identical times on different days.
- Modeled courses, prerequisites, sections, and meetings with Flyway migrations; used JPA fetch joins to load scheduling data in one query and centralized REST validation and error responses.

Card Duel — Turn-Based Strategy Game

C#, Unity

- Built a solo C#/Unity strategy game with a 6x6 exploration grid and turn-based combat; composed card effects from ScriptableObjects while separating content definitions from mutable game state.
- Validated card costs and targets before executing effects; delayed discarding to prevent drawing the active card again, and tracked status activation and expiry across turn boundaries.
- Implemented JSON save/load with checksums, atomic file replacement, and state validation; preserved combat hands, random-generator state, and reward flags to prevent duplicate claims after reload.

The Garbage Man — 3D Top-Down Roguelike

C#, Unity | GitHub | Demo

- Developed and released a 3D roguelike in Unity; composed enemies from interchangeable movement and weapon prefabs to vary behavior while reusing shared health and damage logic.
- Implemented a persistent run manager with PlayerPrefs checkpoints to restore level progression, health, currency, and upgrades across scene changes and subsequent play sessions.

Heartline — UX Case-Study Website

JavaScript, HTML, CSS | Live | GitHub

- Published a responsive UX case-study site for a student friendship prototype, organizing research, eight prototype screens, Figma embeds, and usability findings with GitHub Pages deployment.

EXPERIENCE

Freelance Web Developer — Fiverr

Remote

Independent Client Projects

Jan 2024 – Sep 2024

- Developed responsive websites for businesses and nonprofits using HTML, CSS, JavaScript, WordPress, and Shopify; gathered requirements and incorporated client feedback through delivery.

Innospire Solutions Inc.

Queens, NY

Information Technology Intern

Sep 2023 – Dec 2023

- Supported requirements analysis and technical specifications for business applications; contributed to implementation, testing, and troubleshooting to address operational software needs.